IPS Veto Position Scan for Deep Inelastic Event Tagging

Geant4 study in the 3 deg, 1.15 T configuration

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Physics Goal

- The IPS detector is used as a veto counter for deep inelastic deuteron reactions.
- The optimization target is:
  - maximize the IPS response to small-impact-parameter events,
  - while suppressing the response to elastic deuteron breakup events.
- The study reported here is the current **official full-statistics run**:
  - geometry: `3deg_1.15T`,
  - target: `reference only` (`Target/SetTarget false`),
  - selection: `no-forward`,
  - signal proxy: all-event samples with $b_{\mathrm{imp}} \leq 7$.

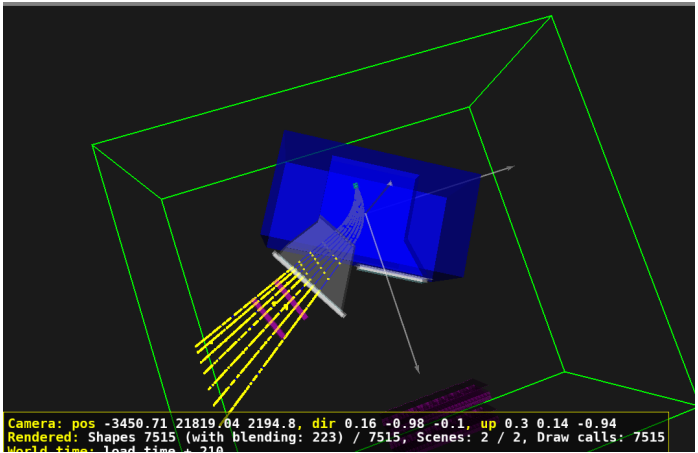
Detector and Coordinate Definition

- IPS is implemented as a 32-channel barrel in Geant4.
- Geometry parameters:
  - 32 modules,
  - module size: $20 \times 20 \times 200$ mm^3,
  - barrel radius: 135 mm,
  - angular pitch: $360/32 = 11.25°$.
- The target normal defines the local beam direction.
- The IPS barrel axis is parallel to this local beam direction.
- The scanned parameter is the signed axial displacement:

$$\vec{r}_{\mathrm{IPS}} = \vec{r}_{\mathrm{target}} + s\,\hat{u}_{\mathrm{beam}},$$

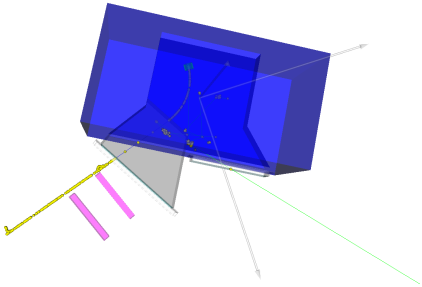
where s is the `AxisOffset`.

Proton px Scan at Offset 0 mm



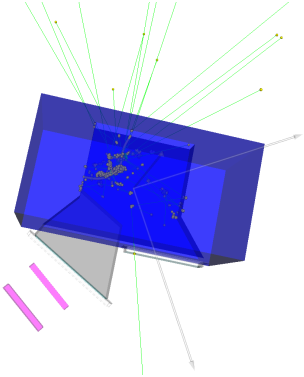
- One event contains 7 injected protons.
- Input family:
- px = [+150, +100, +50, 0, -50, -100, -150] MeV/c
- py = 0 MeV/c, pz = 627 MeV/c
- All tracks start from the same target-reference point.
- The apparent vertical spread in the rendered view comes from projection and magnetic bending, not from an initial y dispersion.

Elastic Breakup Representative at Offset 0 mm



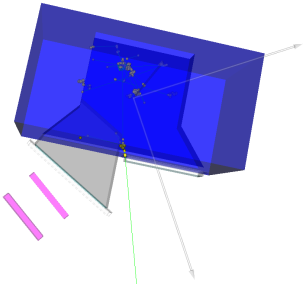
- This picture uses the selected elastic representative from bucket b05.
- Event metadata:
  - $b_{\mathrm{imp}} = 5$
  - 2 final fragments
  - 1 charged fragment
  - IPS hit at 0 mm: no
- Viewpoint: from $+y$ toward the x-z plane.
- It is close to the median relative-momentum topology inside the cleaned elastic sample.
- These visual examples use **offset 0 mm** only for consistent illustration; the optimization itself is still

Small-b Representative Event at Offset 0 mm: Central Topology



- Representative all-event entry from bucket b01.
- Event metadata:
  - $b_{\mathrm{imp}} = 1$
  - 28 final fragments
  - 6 charged fragments
  - IPS hit at 0 mm: yes
- Viewpoint: from $+y$ toward the x-z plane.
- This is the kind of high-multiplicity central event that the veto should detect efficiently.

Small-b Representative Event at Offset 0 mm: Edge of the Signal Region



- Representative all-event entry from bucket b07.
- Event metadata:
  - $b_{\mathrm{imp}} = 7$
  - 11 final fragments
  - 2 charged fragments
  - IPS hit at 0 mm: yes
- Viewpoint: from $+y$ toward the x-z plane.
- This event sits near the upper boundary of the small-b signal definition and illustrates the lower-multiplicity end of the accepted topology family.

Important Modeling Choice

Target is a reference point, not a material

- The primary particles already come from ImQMD post-reaction outputs.
- Therefore the target must **not** be inserted as a real material in Geant4.
- The run macros force:

```
/samurai/geometry/Target/SetTarget false
/samurai/geometry/IPS/SetIPS true
/samurai/geometry/IPS/AxisOffset <offset> mm
```

- The target position and angle are still used to define the launch point and the IPS alignment axis.

Input Event Preparation

Elastic background

- Source: cleaned `dbreak` inputs.
- Unphysical breakup rejection:
  - $|p_y^p - p_y^n| < 150$ MeV/c for y-polarized random-ϕ data,
  - $|p_z^p - p_z^n| < 150$ MeV/c for z-polarized data.
- Full-statistics total:

$$N_{\text{elastic}} = 19305.$$

Signal proxy

- Source: merged `geminiout` all-event inputs.
- Small-b definition:

$$b_{\text{imp}} \leq 7.$$

- No forward gate is applied.
- Full-statistics total:

$$N_{\text{small-}b} = 140000.$$

Why no forward gate?

The IPS is optimized as a veto detector for deep inelastic reactions themselves, not only for events that survive a downstream trigger chain.

Merged Datasets Used in the Scan

Input class	Buckets used	Full-statistics entries
Elastic `dbreak`	`b01` – `b10`	19305
All-event `geminiout`	`b01` – `b07`	140000

- The merged inputs are built from:
  - `d+Sn124-ypolE190g050ypn`
  - `d+Sn124-ypolE190g050ynp`
- Each bucket is merged with `hadd` before Geant4 simulation.
- In the full-statistics run, `beamOn = 0`, which means that every merged ROOT tree is processed completely.

Scan Workflow

1. Convert and merge ImQMD-derived ROOT inputs.
2. For each IPS offset, generate a Geant4 macro with `Target/SetTarget false`.
3. Run `sim_deuteron`.
4. Read back `FragSimData` and count IPS hits.
5. Compute:
   - elastic leakage,
   - small-b IPS acceptance.
6. Rank candidates with the rule:
   1. require `elastic_leakage < 0.005`,
   2. maximize `smallb_selected_rate`,
   3. break ties with lower leakage,
   4. then smaller $|\text{offset}|$.

Scan Grid

- Coarse grid:

$$-200, -160, -120, -80, -40, 0, 40, 80, 120, 160, 200 \text{ mm}$$

- Refinement:
  - take the top-2 coarse candidates,
  - scan ± 20 mm around each center,
  - with 5 mm steps.
- The final full-statistics summary contains all evaluated offsets, not only the shortlist.
- Runtime progress is recorded per offset in:
  - `scan_progress.log`
  - `ips_scan_summary.csv`

Metric Definitions

Elastic leakage

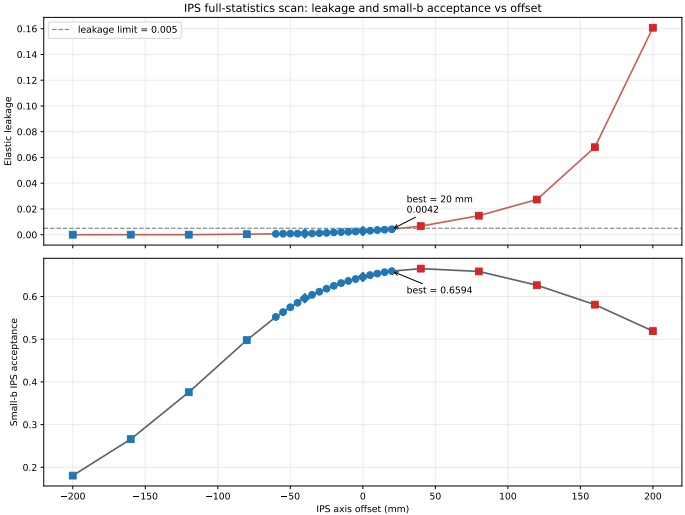
$$\text{leakage} = \frac{N(\text{elastic events with IPS hit})}{N(\text{elastic events})}$$

Small-b IPS acceptance

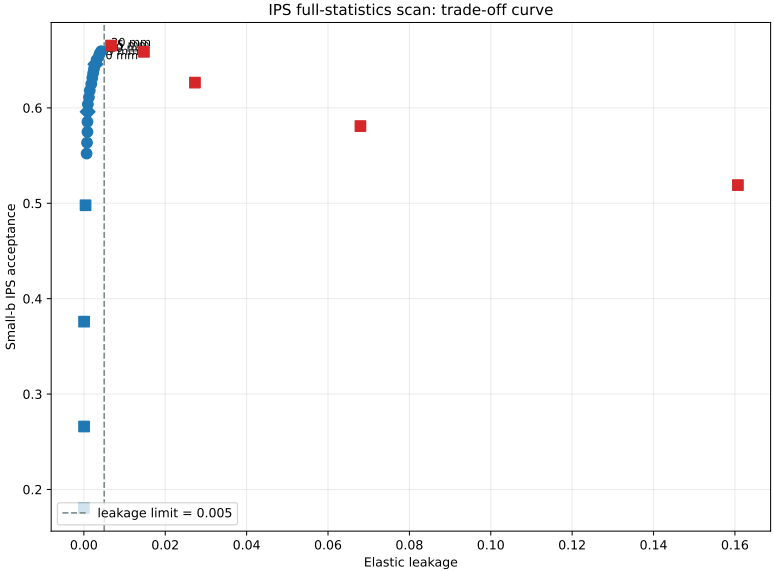
$$\text{acceptance} = \frac{N(\text{all-event entries with } b_{\text{imp}} \leq 7 \text{ and IPS hit})}{N(\text{all-event entries with } b_{\text{imp}} \leq 7)}$$

- In this no-forward setup, `smallb_selected_rate` and `smallb_raw_rate` are identical.
- The feasibility threshold is fixed at 0.5%.

Full-Statistics Result: Metrics vs Offset



Full-Statistics Result: Trade-off Curve



Best Candidates

Rank	Offset (mm)	Feasible	Elastic leakage	Small-b acceptance
1	20	yes	0.004248	0.659436
2	15	yes	0.003885	0.656993
3	10	yes	0.003626	0.653664
4	5	yes	0.003108	0.650193
5	0	yes	0.002849	0.645771

- The official full-statistics recommendation is **+20 mm**.
- Positive offsets above 40 mm become infeasible because elastic leakage exceeds the 0.5% requirement.
- The previous sampled `beamOn300` run favored `-40 mm`; the full-statistics run resolves this finite-statistics bias.

Interpretation

- Moving IPS from 0 mm to +20 mm improves small-b acceptance from:

$$0.645771 \rightarrow 0.659436$$

  while still satisfying the leakage constraint.
- The gain is modest but systematic across the final refined positive offsets.
- Large positive shifts increase the overlap with elastic background too quickly.
- Large negative shifts remain very clean, but lose deep-inelastic coverage.

Artifacts and Reproducibility

Result directory

- `data/simulation/ips_scan/.../smallb_all_noForward_fullstats`

Main outputs inside the result directory

- `best_ips_positions.txt`
- `ips_scan_summary.csv`
- `scan_progress.log`

Driver script

- `run_ips_position_scan_3deg115T_smallb_all_noforward_fullstats.sh`

Plotting script

- `scripts/analysis/plot_ips_scan_metrics.py`

Take-away

Recommendation

For the current Geant4 configuration (`3deg_1.15T`, target-reference-only, no-forward, full statistics), the preferred IPS position is:

`AxisOffset = +20 mm`

- Elastic leakage: 0.4248%.
- Small-b acceptance: 65.94%.
- This is the result to use as the current reference for further veto studies and geometry exports.